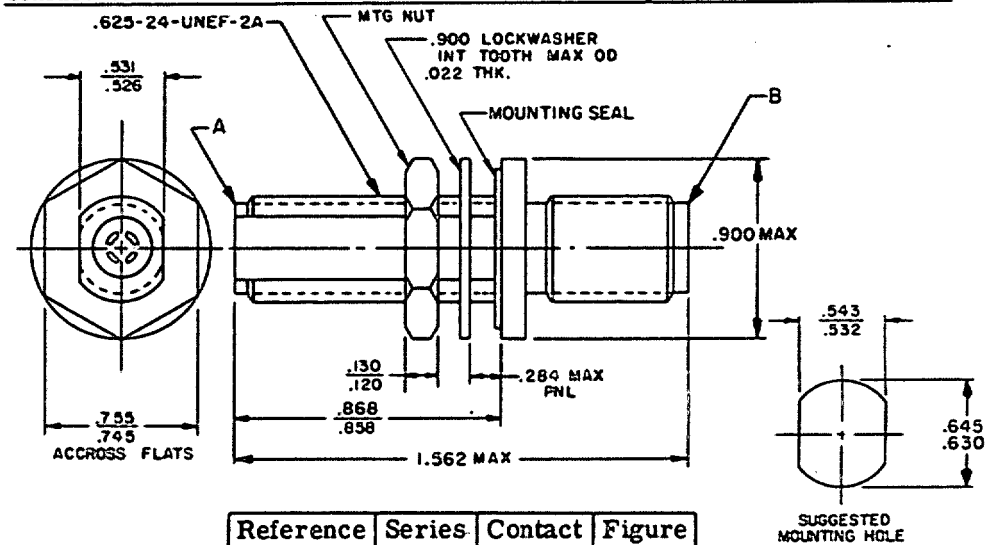
MIL-A-55339/4
6 May 1975
SUPERSEDING
MS90159D
25 February 1966

MILITARY SPECIFICATION SHEET

ADAPTER, CONNECTOR, COAXIAL, RADIO FREQUENCY, (WITHIN SERIES N (HERMETIC)), CLASS 2, STRAIGHT RECEPTACLE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the adapter described herein shall consist of this document and the latest issue of Specification MIL-A-55339.



Reference	Series	Contact	Figure
A & B	N	Socket	2

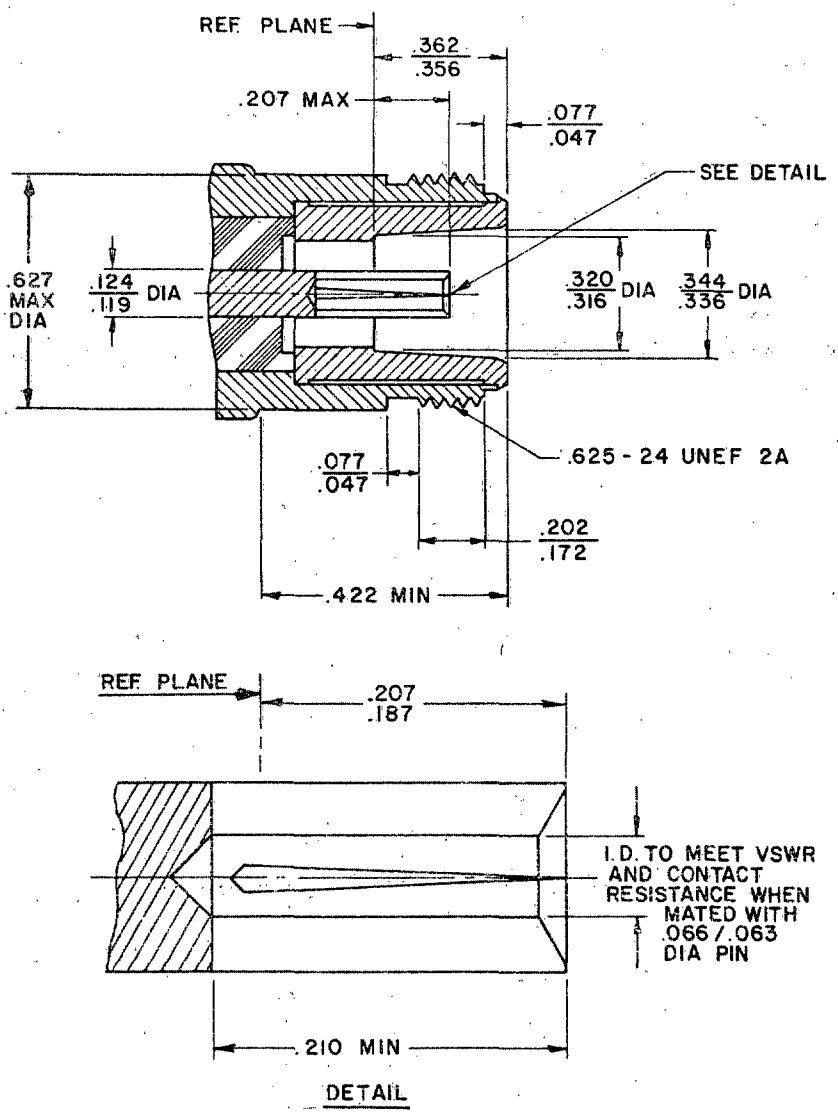
INCHES	MM	INCHES	MM
.020	.51	.630	16.00
.120	3.05	.645	16.38
.130	3.30	.745	18.92
.284	7.21	.755	19.18
.526	13.36	.858	21.79
.531	13.49	.868	22.05
.532	13.51	.900	22.86
.543	13.79	1.562	39.67
.625	15.88		

NOTES:
1. Dimensions are in inches.
2. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
3. All undimensioned pictorial representations are for reference purposes only.

FIGURE 1. General configuration.

FSC 5935

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INCHES	MM
.047	1.19
.063	1.60
.066	1.68
.077	1.96
.119	3.02
.124	3.15
.172	4.37
.187	4.75
.202	5.13
.207	5.26
.316	8.03
.320	8.13
.336	8.53
.344	8.74
.356	9.04
.362	9.19
.422	10.72
.625	15.88
.627	15.93

NOTES:
1. Dimensions are in inches.
2. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
3. Slitting of inner contact optional.
4. All undimensioned pictorial representations are for reference purposes only.

FIGURE 2. Mating dimensions for socket contact terminations.

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DESIGN AND CONSTRUCTION:

General configuration: See figure 1.

Impedance: 50 ohms, nom.

Working voltage: Sea level - 1,000 Vrms.
70,000 feet - 250 Vrms.

Frequency range: 0 to 11 GHz.

Temperature range: -65° to +165°C.

PERFORMANCE (installation torque of 6 to 10 in. lb).

Dimensions: See figures 1 and 2.

Center contact retention: Axial force - 6 lb, min.
Torque - Not applicable.

Force to engage and disengage: Longitudinal force - Not applicable.
Torque - 6 in. lb, max.

Coupling proof torque: Not applicable.

Mating characteristics:
Center contact (socket) open position:
Oversize test pin dia - .074 in., min.
Insertion depth - .125 in., min.
No. of insertions - 1.
Max test pin (insertion force test) closed position:
Steel test pin dia - .066 in., min.
Pin finish - 16 microinches.
Insertion force - 2 lb, max.
No. of insertions - 1.
Min test pin (withdrawal force):
Steel test pin dia - .063 in., max.
Pin finish - 16 microinches.
Withdrawal force - 2 oz, min.
No. of withdrawals - 1.
Outer contact: Not applicable.

Permeability: <2.0.

Seal: Hermetic - 1×10^{-7} atm cc/sec, max (center contact to body).
Pressurized - Not applicable.
Weatherproof - 30 psi, max (mounting seal).

Insulation resistance: 5,000 megohms, min.

VSWR: 1.45:1 max at .5 to 11 GHz.

RF leakage (total): -90 dB, min, 2 to 3 GHz.

RF insertion loss: .2 dB, max, 9 GHz
(.067 $\sqrt{F\ (GHz)}$ dB max tested at 9 GHz).

Durability: 500, min.
Rate: 12 c/m, min.

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Dielectric withstanding: Test voltage - 2,500 Vrms, min (sea level).

Contact resistance (milliohms, max):

Contact	Initial	After
Center	2.5	3.0
Outer	0.3	N/A

Vibration, high frequency: Interruptions - 1 µs, max.

Shock: Test condition I.

Thermal shock: Test condition C.

Moisture resistance: 200 megohms, min.

Corona level: Voltage - 500 V, min.
Altitude - 70,000 feet, min.

RF high potential withstanding voltage: RF voltage - 1,500 Vrms, min.
Frequency - 5 MHz, min.

Salt spray (corrosion): Applicable.

Coupling mechanism retention force: Not applicable.

MARKING: As specified in MIL-A-55339.
Part No. M55339/04-00030.

TABLE I. Cross reference of part numbers.

Part number	Superseded part number or type designation 1/
M55339/04-00030	MS90159 UG-30E/U

1/ The superseded part number or the type designation is for cross reference only. Where a superseded part number or type designation is not given, none was assigned or will be assigned. The part number M55339/04-00030 shall be used in all cases for marking and identifying the adapter.

Custodians:
Army - EL
Navy - EC
Air Force - 85

Review activities:
Army - MU, MI, EL
Navy - SH
Air Force - 11, 80
DSA - ES

User activities:
Army - AT
Navy - AS, MC
Air Force - 19

Preparing activity:
Army - EL

Agent:
DSA - ES

(Project 5935-1918-2)

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